PROJECT 2 • DEDICATED REPORT

IGN Video Game Review Patterns & Platform Analytics

Data cleaning, annual score trajectories, genre distributions, and editorial sentiment modeling.

Dataset: `ign.csv` (18,625 Reviews) **Span:** 1996 to 2016 **Tech:** Pandas, NumPy, Matplotlib

1. Scenario Breakdown & Findings

Scenario 1: Data Preprocessing & Type Enforcement

Dropped redundant index `Unnamed: 0`. Imputed missing score values with mean (μ = 6.95) and genres with mode ('Action'). Cast temporal columns to integers.

Scenario 2: Annual Average Score Trajectory

Grouped by `release_year` and calculated average score via NumPy arrays. Peak average critical score occurred in **1996 (Score: 7.70)**. Line plot: `avg_score_trend.png`.

Scenario 3: Platform Quality Filtering (Score > 7)

Filtered 11,359 high-rated titles (score > 7). Top platforms: **PlayStation 2 (1,141)**, **Xbox 360 (1,131)**, and **PlayStation 3 (914)**. Bar plot: `top_platforms_bar.png`.

Scenario 4: Top 5 Genre Distribution

Aggregated top genres: Action (3,797, 36.1%), Sports (1,916, 18.2%), Shooter (1,610, 15.3%), Racing (1,228, 11.7%), Adventure (1,175, 11.2%). Pie chart: `genre_distribution.png`.

Scenario 5: Advanced Tiering, NumPy Diff & Editorial Correlation

Created `score_category` (*Excellent*: 1,602; *Good*: 9,757; *Average*: 7,266). Pearson correlation between score and Editor's Choice is **0.569** (strong positive relationship). Visuals: `score_trend.png`, `score_category_stacked.png`, `score_distribution.png`.

2. Visual Representation

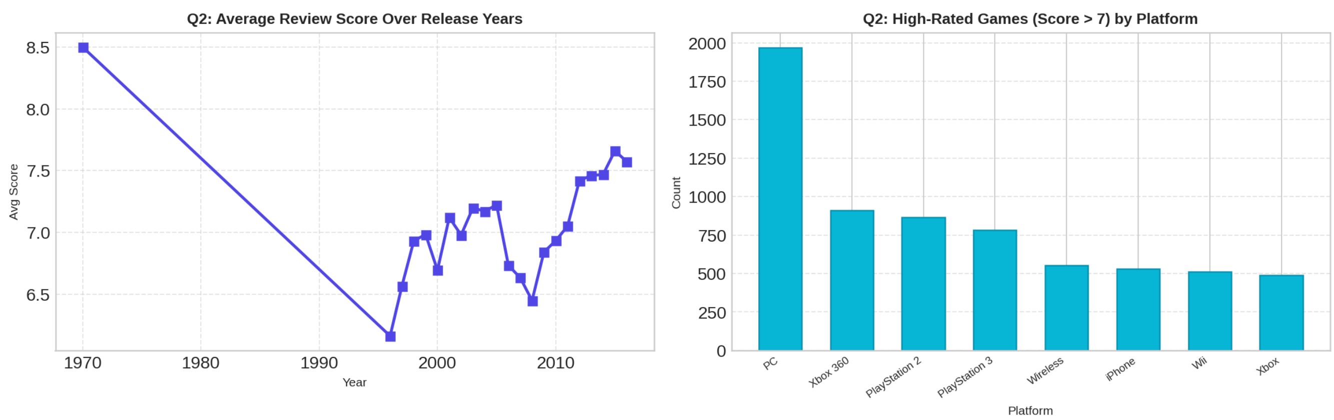

Figure 2.1: Mean Annual Review Score Trend (Left) & High-Rated Game Counts by Platform (Right).

3. Python Code Snippet

```
import os, pandas as pd, numpy as np, matplotlib.pyplot as plt
df = pd.read_csv("ign.csv")
if "Unnamed: 0" in df.columns: df.drop(columns=["Unnamed: 0"], inplace=True)
df["score"] = df["score"].fillna(df["score"].mean())
df["genre"] = df["genre"].fillna(df["genre"].mode()[0])
df["score_category"] = df["score"].apply(lambda s: "Excellent" if s>=9 else ("Good" if s>=7 else "Average"))
print(f"Top Platform (Score > 7): {df[df['score']>7]['platform'].value_counts().idxmax()}")
```